

Digital Literacy Foundations

Course overview

Target age: 7–12

Length: 12 lessons

Suggested lesson duration: 45–60 minutes

Environment: Controlled virtual desktop with a lesson-specific simulated web

Primary focus: Practical use of computers rather than programming

The course develops four connected areas:

1. Keyboard and mouse use
2. Window and application management
3. File and folder management
4. Web navigation, searching and information evaluation

The current National Curriculum for England expects primary pupils to use technology purposefully; create, organise, store, manipulate and retrieve digital content; use search technologies effectively; evaluate digital content; and use technology safely and responsibly. This course concentrates on those information-technology and digital-literacy requirements rather than the computer-science elements of the computing curriculum.

As of August 2026, these remain the relevant statutory computing programmes of study. The revised national curriculum is planned for first teaching from September 2028.

Course aims

By the end of the course, pupils should be able to:

- control a computer accurately using a mouse and keyboard
- open, close, move, resize and switch between application windows
- enter, select, edit, copy, cut and paste text
- understand the difference between files, folders and applications
- create, save, name, locate and reopen files
- organise files using folders
- move, copy, delete and restore files
- navigate a web browser using links, tabs and standard controls
- construct useful search queries
- distinguish between relevant, irrelevant and promoted search results
- make basic judgements about whether online information is trustworthy
- download a file and organise it within the virtual file system
- combine several digital skills to complete a practical task independently

Course structure

Unit	Lessons	Main focus
Computer control	1–4	Mouse, keyboard, editing and windows
Files and folders	5–7	Saving, locating and organising work
Using the web	8–11	Browsing, searching, evaluating and downloading
Consolidation	12	Connected practical skills challenge

Unit 1: Computer control

Lesson 1 — Controlling the Computer

Learning objectives

Pupils will:

- recognise the main parts of the virtual desktop
- distinguish between the pointer and text cursor
- click, double-click, right-click, drag and scroll
- open and close an application

Key vocabulary

Desktop, icon, pointer, cursor, click, double-click, right-click, drag, scroll, application

Lesson activities

1. Identify the desktop, application icons, Start-style menu and taskbar.
2. Complete a pointer-control activity involving targets of different sizes.
3. Drag objects into the correct positions.
4. Scroll through a document or image.
5. Open an application from the desktop.
6. Close the application using its window control.

Practical check

The pupil must independently:

- open a named application
- interact with an object inside it

- scroll to a specified point
- close the application

Extension

Use a right-click menu to complete an action.

Lesson 2 — Keyboard Confidence

Learning objectives

Pupils will:

- locate letters, numbers and common punctuation
- use Space, Enter, Backspace and Shift
- enter capital letters correctly
- move through text using the mouse and arrow keys
- correct errors in existing text

Key vocabulary

Keyboard, key, text cursor, Space, Enter, Backspace, Shift, arrow keys

Lesson activities

1. Complete a keyboard scavenger hunt.
2. Type a short prepared sentence.
3. Add capital letters and punctuation.
4. Correct deliberate spelling and spacing errors.
5. Move the text cursor without deleting existing text.
6. Insert missing words into a paragraph.

Practical check

The pupil reproduces a short passage containing:

- capital letters
- punctuation
- numbers
- a new line

Extension

Navigate through the text using Home, End and the arrow keys.

Lesson 3 — Selecting, Copying and Pasting

Learning objectives

Pupils will:

- select letters, words and larger sections of text
- copy and paste text
- cut and move text
- undo an unwanted change
- understand the difference between copying and moving content

Key vocabulary

Select, highlight, copy, cut, paste, clipboard, undo

Lesson activities

1. Select a word, sentence and paragraph.
2. Copy a repeated phrase rather than typing it again.
3. Reorder a mixed-up paragraph using cut and paste.
4. Copy information from one part of a document to another.
5. Make an intentional mistake and use Undo to correct it.
6. Repeat the tasks using keyboard shortcuts.

Core shortcuts

- **Ctrl+A** — Select all
- **Ctrl+C** — Copy
- **Ctrl+X** — Cut
- **Ctrl+V** — Paste
- **Ctrl+Z** — Undo

Practical check

The pupil must:

1. Select the correct text.
2. Copy it into a specified location.
3. Cut a different section.
4. Paste it into the correct position.
5. Undo one deliberate error.

Extension

Complete the task without using toolbar buttons.

Lesson 4 — Managing Windows

Learning objectives

Pupils will:

- recognise the main parts of an application window
- move and resize windows
- minimise, maximise and restore windows
- switch between open applications
- arrange two applications so they can be used together

Key vocabulary

Window, title bar, taskbar, minimise, maximise, restore, resize, switch

Lesson activities

1. Open two applications.
2. Move one window by dragging its title bar.
3. Resize a window using its edges or corners.
4. Minimise and restore an application.
5. Maximise and restore an application.
6. Switch between applications using the taskbar.
7. Arrange instructions beside a Text Editor window.

Practical check

The pupil receives a short window-management mission:

Open the instructions and Text Editor. Place them side by side, minimise the instructions, restore them and then close both applications.

Extension

Use **Alt+Tab** to switch between applications.

Unit 2: Files and folders

Lesson 5 — Understanding Files and Folders

Learning objectives

Pupils will:

- explain the difference between a file and folder
- recognise that applications create different file types
- navigate through folders
- use folder names and file icons to predict their contents
- understand that a file has a name and location

Key vocabulary

File, folder, location, document, image, file type, path

Lesson activities

1. Sort example objects into files and folders.
2. Explore a prepared folder structure.
3. Open folders and return to the previous location.
4. Identify text files and image files.
5. Match files to the applications that open them.
6. Locate named files within nested folders.

Practical check

The pupil must locate three specified files in a prepared folder structure.

Extension

Follow a written path such as:

My Files > School Work > Animals > Images

Lesson 6 — Saving and Reopening Work

Learning objectives

Pupils will:

- save a new file
- choose an appropriate filename

- select the correct folder
- reopen saved work
- understand the purpose of Save As

Key vocabulary

Save, Save As, filename, location, overwrite, version

Lesson activities

1. Create a short document in the Text Editor.
2. Open the Save window.
3. choose the correct folder.
4. Enter a meaningful filename.
5. Save and close the document.
6. Locate and reopen the file using File Manager.
7. Use Save As to create a second version.

Filename examples

Unclear filename	Improved filename
document1	Rainforest Notes
stuff	Space Facts
picture	Blue Robot Drawing
final2	Animal Poster Revised

Practical check

The pupil must create, save, close, locate and reopen a file without assistance.

Extension

Explain what should happen when a filename is already in use.

Lesson 7 — Organising and Recovering Files

Learning objectives

Pupils will:

- create and rename folders
- move files using drag-and-drop and cut-and-paste
- copy files

- delete files
- restore deleted files from the Recycle Bin
- search for a file

Key vocabulary

Create, rename, move, copy, delete, Recycle Bin, restore, search

Lesson activities

Pupils receive a deliberately untidy folder containing:

- images mixed with documents
- unclear filenames
- duplicate files
- files in incorrect folders
- one apparently missing file

They must:

1. Create appropriate folders.
2. Rename unclear files.
3. Move files into the correct folders.
4. Copy a file where a second copy is required.
5. Delete an unnecessary duplicate.
6. Restore a file from the Recycle Bin.
7. Use search to locate another file.

Practical check

The final file structure is checked automatically against the required state.

Extension

Organise the same files in a different sensible way and explain the choices made.

Unit 3: Using the web

Lesson 8 — Using a Web Browser

Learning objectives

Pupils will:

- recognise the main parts of a browser

- distinguish between a browser, website and webpage
- use the address bar
- follow links
- use Back, Forward, Refresh and Home
- open, switch between and close tabs

Key vocabulary

Internet, web, browser, website, webpage, address bar, link, tab

Lesson activities

1. Identify the address bar and browser controls.
2. Enter a prepared web address.
3. Follow links through a simulated website.
4. use Back and Forward to retrace a route.
5. Refresh a page.
6. Open links in separate tabs.
7. Switch between tabs and close unwanted ones.

Practical check

The pupil must navigate to a named page, open a second page in another tab and return to the original page.

Extension

Use:

- Ctrl+L
- Ctrl+T
- Ctrl+W

Lesson 9 — Searching Effectively

Learning objectives

Pupils will:

- identify keywords in a question
- construct a useful search query
- improve a search that produces poor results
- scan result titles and descriptions
- recognise promoted or sponsored results

Key vocabulary

Search engine, keyword, search query, result, relevance, sponsored result

Lesson activities

1. Compare broad and specific searches.
2. Identify the important words in a research question.
3. Improve deliberately weak searches.
4. Compare results generated by different queries.
5. Identify relevant and irrelevant results.
6. Distinguish between normal and sponsored results.

Examples

Broad search	Improved search
jaguar	jaguar animal habitat
space	distance from Earth to Mars
Romans	what did Roman children eat
sharks	how long do great white sharks live

Practical check

The pupil must use the simulated search engine to find three specific pieces of information using suitable queries.

Extension

Refine a search using an additional keyword or exact phrase.

The Key Stage 2 curriculum requires pupils to use search technologies effectively, understand that results are selected and ranked, and be discerning when evaluating digital content.

Lesson 10 — Can We Trust This Website?

Learning objectives

Pupils will:

- understand that online information is not automatically accurate
- identify who created a webpage
- recognise the likely purpose of a page
- distinguish fact, opinion, advertising and unsupported claims

- compare information across more than one source

Key vocabulary

Source, author, evidence, fact, opinion, advert, reliable, misleading

Lesson activities

Pupils compare prepared simulated websites about the same subject. These might include:

- a museum or educational website
- a personal blog
- a fictional news page
- an advert disguised as an article
- a page containing exaggerated or unsupported claims

Pupils use a simple checking framework.

STOP check

- **S — Source:** Who created it?
- **T — Time:** When was it published or updated?
- **O — Objective:** Why was it created?
- **P — Proof:** What evidence is provided?

Practical check

The pupil selects the more dependable of two sources and identifies evidence supporting the decision.

The Education for a Connected World framework describes age-related knowledge and skills for navigating online technology, including managing online information and recognising how digital content can influence behaviour.

Lesson 11 — Downloading and Using Files

Learning objectives

Pupils will:

- recognise an appropriate download link
- download a prepared file safely
- locate the downloaded file
- rename and move it
- open it in the appropriate application
- use information from one application in another

Key vocabulary

Download, Downloads folder, file type, open, move, source

Lesson activities

1. Search for a required image or document.
2. Select an appropriate result.
3. Download the prepared file.
4. Open the Downloads folder.
5. Rename the downloaded file.
6. Move it into a project folder.
7. Open it in Image Viewer.
8. Copy selected text from the browser into a document.
9. Save the document in the same project folder.

Practical check

The pupil must download, rename, organise and open a specified file.

Extension

Locate and correct a file downloaded into the wrong folder.

Unit 4: Consolidation

Lesson 12 — Digital Skills Challenge

This lesson is a sequence of connected practical tasks rather than an open-ended project.

Learning objectives

Pupils will:

- select appropriate digital skills without being told every individual step
- combine window, file and browser skills
- recover from common mistakes
- demonstrate increasing independence

Challenge scenario

The pupil opens a prepared desktop containing:

- a set of instructions
- several disorganised files

- a partially completed document
- an item in the Recycle Bin
- access to a lesson-specific simulated web
- a required downloadable image

Required tasks

The pupil must:

1. Open the instructions.
2. Open the Text Editor.
3. Arrange the two windows side by side.
4. Open the supplied work folder.
5. Create the required subfolders.
6. Rename two unclear files.
7. Move the files into the correct folders.
8. Restore a required image from the Recycle Bin.
9. Use the simulated search engine to find a specified fact.
10. Select the more reliable of two websites.
11. Copy the required information into the document.
12. Download a suitable image.
13. Locate it in the Downloads folder.
14. Rename and move it into the correct project folder.
15. Open the image in Image Viewer.
16. Save the completed document using the specified name and location.

Assessment

The platform checks the resulting desktop state and the actions completed.

The assessment should distinguish between:

- completed independently
- completed after a general hint
- completed after a specific hint
- completed with tutor assistance
- not yet completed

This final lesson measures practical digital independence without making writing ability, artistic ability or subject knowledge the main focus.

Teaching approach

Suggested lesson structure

Stage	Time	Purpose
Retrieval activity	5 minutes	Revisit a previous practical skill
Demonstration	5–10 minutes	Introduce the new concept
Guided activity	10 minutes	Complete an example together
Independent task	20–25 minutes	Apply the skill
Review	5 minutes	Complete a practical check or explain a decision

Progression

The course should gradually reduce support:

1. **Early lessons:** Controls are highlighted and instructions are highly specific.
2. **Middle lessons:** Pupils follow written task sequences independently.
3. **Later lessons:** Pupils are given an outcome and decide which controls to use.
4. **Final lesson:** Several skills are combined in a prepared scenario.

Mixed-age differentiation

The course covers ages 7–12, so difficulty should be based on independence and complexity rather than assigning completely different lesson topics.

Supported level

- visual instructions
- fewer files and folders
- larger click targets
- controls highlighted when needed
- one application used at a time
- simple filenames
- immediate feedback

Core level

- written instructions
- multiple open applications
- nested folders

- independent use of menus
- meaningful filenames
- simple website comparisons

Extension level

- keyboard shortcuts
- reduced guidance
- more complex folder structures
- ambiguous or irrelevant search results
- alternative valid solutions
- explanation of decisions
- recovery from deliberately introduced mistakes

Curriculum mapping

National Curriculum expectation	Course coverage
Use technology purposefully	All lessons
Create digital content	Text Editor and Paint activities
Organise and store digital content	Lessons 5–7 and 11–12
Manipulate and retrieve digital content	Lessons 3 and 5–7
Use search technologies effectively	Lesson 9
Understand how results are selected and ranked	Lesson 9
Be discerning when evaluating digital content	Lesson 10
Use technology safely and responsibly	Lessons 8–11
Select and combine applications	Lessons 4, 11 and 12
Recognise acceptable and unacceptable behaviour	Website evaluation and safe-download activities

Platform Requirements

1. Virtual desktop

The course requires a neutral, browser-based desktop environment containing:

- desktop area
- application icons

- Start-style application launcher
- taskbar
- clock
- open-application indicators
- Recycle Bin
- active and inactive window states

It should use familiar desktop conventions without directly copying Windows, macOS or ChromeOS.

Each task must be able to decide which desktop features and applications are visible.

2. Window system

Application windows must support:

- opening
- closing
- moving
- resizing
- minimising
- maximising
- restoring
- overlapping
- focusing
- switching through the taskbar
- arranging two windows side by side

Optional extensions include:

- `Alt+Tab`
 - window snapping
 - keyboard-controlled window switching
-

3. Keyboard and mouse interactions

The environment must recognise:

- click
- double-click
- right-click
- click-and-drag
- drag-and-drop
- scrolling
- text entry

- text selection
- cursor movement
- keyboard shortcuts

The platform should distinguish between interaction methods where relevant. A task could require a pupil to use drag-and-drop, a context menu or a keyboard shortcut.

4. Shared clipboard

A common clipboard must work across supported applications.

It needs to support:

- copying text
- cutting text
- pasting text
- copying files
- cutting and moving files
- pasting files into folders
- right-click clipboard commands
- keyboard clipboard shortcuts
- Undo after text editing

It should work at minimum between:

- Text Editor and Text Editor
 - Browser and Text Editor
 - File Manager folders
-

5. Virtual file system

The file system must support:

- files
- folders
- nested folders
- filenames
- file types
- file locations
- created and modified states
- a Downloads folder
- a Recycle Bin
- task-prepared starting files
- optional state carry-over between tasks

Required file operations:

- create
- open
- save
- Save As
- rename
- move
- copy
- delete
- restore
- search
- sort

Applications must use the same file system so that a file created in one application can be found and opened in another.

6. File Manager

The File Manager needs:

- folder and file views
- recognisable file-type icons
- folder navigation
- Back navigation
- breadcrumb or path display
- New Folder
- rename
- drag-and-drop
- cut, copy and paste
- delete
- Recycle Bin access
- restore
- search
- basic sorting

It does not initially need advanced permissions, cloud synchronisation, compressed files or network drives.

7. Shared Open and Save controls

Applications that create or use files need consistent:

- Open
- Save
- Save As

- filename entry
- folder selection
- New Folder
- file-replacement warning

These controls are important because selecting the correct filename and location is part of the curriculum.

8. Text Editor

The Text Editor needs:

- text entry
- text selection
- cut, copy and paste
- undo
- cursor movement
- basic formatting
- Open
- Save
- Save As

Basic formatting may include:

- bold
- italic
- underline
- font size
- alignment

More advanced word-processing features are unnecessary for this course.

9. Paint application

Paint needs only:

- pencil or brush
- simple shapes
- colour selection
- eraser
- fill
- undo
- Open
- Save
- Save As

Its purpose is to show that applications create different kinds of files and that those files can be located and reopened elsewhere.

Paint is useful but not central to every lesson. It could appear in optional activities or as an alternative file-creation task.

10. Image Viewer

Image Viewer needs:

- open image
- display image
- zoom
- move between images in the same folder
- show the filename
- close

It does not need editing tools.

11. Simulated browser

The simulated browser requires:

- address bar
- Back
- Forward
- Refresh
- Home
- links
- page scrolling
- tabs
- loading state
- prepared websites
- simulated search engine
- controlled downloads

The browser should not access the unrestricted live web.

12. Lesson-specific simulated web

Each lesson should define its own controlled web environment.

A lesson configuration may specify:

- available websites
- website pages
- links between pages
- search terms
- search-result rankings
- relevant results
- irrelevant results
- sponsored results
- misleading pages
- downloadable files
- unavailable or broken pages

This allows the web to be tailored precisely to the intended lesson objective.

For example:

- Lesson 8 can provide a small connected website for navigation practice.
- Lesson 9 can provide search results that react to broad and specific keywords.
- Lesson 10 can provide conflicting sources for reliability checking.
- Lesson 11 can provide approved downloadable files.

13. Simulated search engine

The search system must:

- accept free-text search queries
- match queries against prepared content
- return differently ranked results
- display titles and descriptions
- distinguish sponsored results
- include relevant and irrelevant results
- allow lesson checks to inspect the pupil's query

The search engine does not need to reproduce the complexity of a real search provider. It needs to respond consistently enough to teach keyword selection and result evaluation.

14. Controlled download system

The browser must be able to place prepared files into the pupil's virtual Downloads folder.

Downloads may include:

- images
- text documents
- simple information files

The platform should be able to check:

- which file was downloaded
 - where it was downloaded
 - whether it was renamed
 - whether it was moved
 - whether it was opened in the correct application
-

15. Configurable task environments

Every task should be able to specify:

- visible desktop elements
- available applications
- initial files and folders
- initially open windows
- browser websites
- browser history
- available search results
- downloadable files
- enabled and disabled controls
- whether task state carries forward
- whether the environment resets
- expected final state

A task might expose only the Text Editor in an early lesson, while the final challenge exposes the complete desktop.

16. Carry-over and reset controls

The platform already has a carry-forward setting. This should support:

- carrying files from one task into the next
- carrying application content forward
- retaining the current folder structure
- starting a later task from a prepared snapshot
- resetting an individual task
- resetting the whole lesson
- restoring the original lesson state

This allows a lesson to contain several connected activities without requiring permanent pupil desktops between lessons.

17. Practical task checks

The existing checks system should be expanded to recognise practical desktop actions and resulting states.

Useful checks include:

- application opened
- application closed
- window moved
- window resized
- window minimised
- window maximised
- two windows arranged correctly
- text entered
- text selected
- content copied
- content cut
- content pasted
- folder created
- file saved
- file renamed
- file moved
- file copied
- file deleted
- file restored
- correct file opened
- browser page visited
- browser tab opened
- search query submitted
- correct search result selected
- file downloaded
- correct final folder structure produced

Checks should normally assess meaningful outcomes rather than every low-level movement.

For example, the system should check that a file reached the correct folder, not the exact pointer route used to move it.

18. Hints and staged support

Tasks should support increasingly specific hints:

1. Restate the required outcome.
2. Identify the relevant application or area.
3. Name the control or action required.
4. Highlight the control.

The platform should record which level of support was used so that completion with extensive help can be distinguished from independent completion.

19. Teacher work viewing

Teachers need to be able to:

- view the pupil's current virtual desktop
- inspect open applications
- view the pupil's files and folder structure
- open saved work
- inspect downloaded files
- see completed and incomplete checks
- see hints used
- reset a task
- restore the starting state
- manually mark a task when needed

The platform does not need to record every key pressed. It needs enough visibility to understand the pupil's work and provide support.

20. Accessibility and mixed-age support

The initial platform should provide:

- scalable interface elements
- clear focus indicators
- keyboard navigation
- large click targets where configured
- optional reduced visual complexity
- picture-supported instructions
- optional read-aloud instructions
- adjustable task difficulty

Individual tasks should be able to change:

- number of files
 - depth of folder structures
 - size of controls
 - amount of written guidance
 - number of distracting search results
 - number of applications available
-

Minimum application set

The first course requires:

1. File Manager
2. Text Editor
3. Simulated Browser
4. Image Viewer
5. Recycle Bin

Paint is recommended but could be added after the core environment is functioning.

An Email Simulator is **not required for this 12-lesson course**. It would fit more naturally into a later course focused on online communication, attachments, spam, phishing, privacy and respectful online behaviour.

Minimum platform definition

The course can be delivered with:

A neutral virtual desktop containing movable application windows, a shared virtual file system, File Manager, Text Editor, Image Viewer, Recycle Bin, shared copy-and-paste controls, a lesson-configurable simulated browser and search engine, controlled downloads, configurable task states, practical automated checks and teacher work viewing.

This is substantial enough to teach genuine transferable computer skills without attempting to reproduce an entire operating system.